MILITARY SPECIFICATION

VALVE, LITHIUM BROMIDE STEAM CONTROL, QUIET OPERATION

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers quiet design, air diaphragm operated control valves for the control of steam to a lithium bromide plant.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-V-3 - Valves, Fittings, and Flanges (Except for Systems Indicated Herein); Packaging of.
MIL-R-196 - Repair Parts, Accessories, and Kits, Mechanical; Packaging of.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-R-17131 - Rods and Powders, Welding, Surfacing.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY (Continued)

MIL-G-21032 - Gaskets, Metallic-Asbestos, Spiral Wound.
MIL-I-45208 - Inspection System Requirements.

STANDARDS

FEDERAL

FED-STD-H28/2 - Unified Thread Form and Thread Series For Bolts, Screws, Nuts, Tapped Holes and General Applications.

MILITARY

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-271 - Nondestructive Testing Requirements for Metals.
MIL-STD-278 - Fabrication Welding and Inspection; and Casting Inspection and Repair for Machinery, Piping and Pressure Vessels in Ships of the United States Navy.
MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.
MS16142 - Gasket Seal Straight Thread Tube Fitting, Standard Dimensions for.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A105 - Forgings, Carbon Steel, for Piping Components.
A193 - Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service.
A194 - Carbon and Alloy Steel Nuts for Bolts for High-Pressure and High-Temperature Service.
A216 - Carbon Steel Castings Suitable for Fusion Welding for High-Temperature Service.
A395 - Ferritic Ductile Iron Pressure-Retaining Castings for Use at Elevated Temperatures.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) (Cont'd)
A696 - Steel Bars, Carbon, Hot-Rolled and Cold-Finished, Special Quality, for Pressure Piping Components and Other Pressure-Containing Parts.
B26 - Aluminum-Alloy Sand Castings.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
B1.12 - Class 5 Interference - Fit Thread.
B16.5 - Steel Pipe Flanges and Flanged Fittings.
B16.34 - Steel Butt-Welded End Valves.
B18.2.1 - Square and Hex Bolts and Screws.
B18.2.2 - Square and Hex Nuts.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish a sample valve for first article inspection and approval (see 4.3 and 6.3).

3.2 Valve description. Valves shall be single-seated construction. Flow shall be controlled by the position of the valve plug, which in turn controls the flow path through a quiet-flow regulating element. The valve shall be of air-to-open configuration. Flow through the valve body shall be in the direction of valve plug opening (flow tending to open valve). The diaphragm actuator shall have a 12 pounds per square inch (lb/in^2) span, and shall be capable of operating the valve with a 3 to 15 pounds per square inch gage (lb/in^2g) input signal range on a "bench test" basis (stuffing box handtight and valve body at atmospheric pressure). Due to the variation in steam pressure differential across the valve which can occur in service (see 3.5), a positioner supplied with a 40 lb/in^2 air source will be mounted to the valve.

3.3 Materials. Materials shall be as specified in table I. Materials for parts not listed in table I shall be corrosion-resistant, suitable for the intended pressures and temperatures, and selected to prevent galling, seizing, or excessive wear on operating parts. Clearances shall prevent

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interferences due to the thermal expansion. Materials shall withstand nondestructive tests (NTD), at the option of the Government representative.

TABLE I. List of materials.

Name of part	Materials
Body, bonnet, and bottom cover	ASTM A105 ASTM A216, grade WCB ASTM A696, grade C
Yoke	ASTM A395 ASTM A216, grade WCB ASTM B26, Alloy B443.0 or 443.0
Valve stem and packing follower	Corrosion-resistant steel (300 series)
Valve plug, seat ring, guide posts and bushings	Stellite 1/ Hardened 400 series Corrosion-resistant steel 2/
Bolting	ASTM A193, grade B7 ASTM A194, grade 2H
Gaskets	MIL-G-21032, class B
Packing	See 3.6.1.6.1
Diaphragm	See 3.6.2.5
Diaphragm spring	See 3.6.2.6

1/ Stellite may consist of either wrought stellite 6B, cast stellite 6, or an inlay of stellite (not less than 3/32 inch thickness for main seat and disc surfaces). Where inlays are used, welding rods shall be in accordance with type MIL-RCoCr-A of MIL-R-17131.

2/ Hardened corrosion-resistant steel trim shall consist of a hardened corrosion-resistant steel plug (400 series or 17-4PH) and a stellite (see note 1) seat ring. Guiding surfaces shall be hardened corrosion-resistant steel or stellite (see note 1).

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3.3.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.4 Design.

3.4.1 Design concept. Valves will be operated, maintained, and repaired on board Naval ships and shall emphasize simplicity, reliability, maintainability, and ruggedness. Design shall permit access for adjustment and repair when working from either side of the valve without requiring removal of the valve body from the line.

3.4.2 Interchangeability. Valve design shall permit like parts to be interchangeable between all valves without requiring individual modification of such parts. Each part shall have part number identity and shall be replaceable from stock or the manufacturer on a nonselective and random basis. Parts having the same manufacturer's part number shall be directly interchangeable with each other with respect to installation (physical) and performance (function). Physically interchangeable assemblies, components, and parts are those which are capable of being readily installed, removed, or replaced without alteration, misalinement, or damage to parts being installed or to adjoining parts. No fabrication operations such as cutting, filing, drilling, reaming, hammering, bending, prying, or forcing shall be required. This is not intended to preclude the use of special tools, fixtures, and other shop aids during original assembly of the parts into the article.

3.4.3 Threads. Threads shall conform to FED-STD-H28/2. Where necessary, means shall be provided to prevent the accidental loosening of threaded parts. Pipe threads shall not be used. ANSI B18.2.1 and ANSI B18.2.2 hex-head standards shall be used.

3.5 Reliability. Valve shall demonstrate a mean-time-between-failure (MTBF) by continuous operation of 750 hours without failure. If failure occurs, valve shall be inspected for cause of failure. If correction can be made by adjustment the test may be continued, if not the valve shall be repaired and retested. If failure again occurs, the valve shall be rejected.

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3.6 Construction.

3.6.1 Body subassembly.

3.6.1.1 Pressure-temperature rating. Pressure-temperature rating shall be 600 pound series in accordance with ANSI B16.34.

3.6.1.2 Body. Valve bodies shall be machined from a one-piece casting or forging and be a basic globe configuration with in-line inlet and outlet ports.

3.6.1.3 End preparation. End connections shall be in-line 3 inch iron pipe size (IPS), 600 pound series, raised-face flanges in accordance with ANSI B16.5. Bolt holes shall straddle the vertical centerline. Face-to-face distance between the raised flange faces shall be 15-1/2-inches (plus or minus 1/16-inch). Inlet and outlet bore diameter shall be 2.90 (plus or minus 0.01 inches) and shall blend smoothly with the valve interior flow passages.

3.6.1.4 Bonnet and bottom cover joints. Bonnet and bottom cover (where applicable) shall be flanged for attachment to the body. Joints shall be secured by either of the following:

(a) Through-bolts threaded the entire length and fitted with a nut on each end. Threads on both bolts shall be class 2 fit in accordance with FED-STD-H28/2.
(b) Tap-end studs with interference fit at the tap end and a class 2 fit at the nut end. Interference fit shall be in accordance with ANSI B1.12.

Bonnet and bottom cover shall be located by body guiding by providing a close tolerance fit between machined diameters on the body, bonnet, and bottom cover rather than by depending on studs or bolts for location. Spiral wound gaskets shall be fully retained and the joints shall have metal-to-metal take-up to provide controlled compression of the gaskets. To enable easy gasket removal, not more than two gasket-retaining faces for each gasket shall be formed on a single part. Joint design shall provide for proper alinement of the guide bushings. Bolting area shall be provided to maintain metal-to-metal make-up over at least a 3-year period without retightening. Bearing surface of nuts and their contacting surfaces on the valve shall be finished machined.

3.6.1.5 Bonnet construction. Valves shall limit packing temperatures to a maximum of 500°F under any operating condition. A radiation or extension bonnet shall be provided, where necessary, for this purpose. A packing lubricator shall not be required or used.

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3.6.1.6 Packing box assembly. A packing box assembly shall be provided to seal against leakage along the stem. Sealing bellows shall not be used. Packing gland shall be of bolted construction so that the actuator assembly can be removed without disturbing the packing adjustment.

3.6.1.6.1 Packing. Packing shall be capable of sustaining sealing during valve operation at the temperature and pressure conditions to which it will be subjected.

3.6.1.6.2 Gaskets. Spiral wound gaskets shall retain sufficient residual load in service to maintain a leak-tight joint over at least a 3-year period under operating conditions.

3.6.1.7 Internal trim. Internal trim shall be of cage construction. The quiet-flow regulating element and the seat ring shall be clamped in position by a cage which shall be either a part of, or held in position by, the bonnet or bottom cover flange. This shall permit rapid replacement of all internal trim by avoiding the use of any threads within the valve cavity (such as seat-ring threads). The cage shall remain under sufficient compression at all times to prevent any movement or looseness and the quiet-flow regulating element shall be confined and compressed on both top and bottom by this cage construction.

3.6.1.8 Quiet-flow regulating element. The element used to achieve a low-noise pressure loss shall not incorporate excessively small flow passages or other features jeopardizing operational reliability. Element shall be a disc stack type design, bolted or otherwise fabricated into a one-piece configuration capable of withstanding installation and operational stresses. The method of fabrication shall enable breakdown of the disc stack into individual discs to facilitate cleaning, unless it can be demonstrated that the disc stack can be effectively cleaned without such breakdown. If brazing is permitted and employed, the size and distribution of the bonding surfaces of the individual discs shall be such as to ensure an adequate percentage bond and freedom from flow passage obstruction under all production tolerances. Conformance with this requirement shall be verified by analysis of the fabrication method to be used and by inspection of the elements (see 3.12.1(b) and (c)). Internal passages shall not clog with use of a 0.033 inch diameter perforated metal screen type strainer installed upstream. Elements shall not require cleaning more often than annually. Elements shall not require replacement at less than 5 year intervals due to failures such as blockage, erosion, or wear. The element shall be capable of being cleaned on board ship, and any special equipment or fixtures required for cleaning shall be provided with each valve. The element structure shall be of a solid or continuous nature, not susceptible to media migration.

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3.6.1.9 Guiding surfaces. A difference in hardness of at least 100 points Brinell shall be maintained between the rubbing surfaces for main valve guiding. This requirement shall not apply if both the guide posts and bushings are stellited or if the hardness of either exceeds 450 Brinell. Rubbing surfaces for valve guiding shall have a finish roughness height rating (RHR) of 16 or better.

3.6.2 Diaphragm actuator assembly.

3.6.2.1 Operating range. The valve shall have a bench test span of 12 lb/in^2. The actuator assembly shall be capable of operating the valve on a bench test basis with a standard 3 to 15 lb/in^2 input signal, producing a stem travel which is linear within plus or minus three percent (of full travel), over the full rated stroke of the valve. Due to variations in steam line differential pressure which occurs across the valve in service (see 3.9), a positioner working off a 40 lb/in^2 air source will be mounted to the valve. With the positioner mounted in place and activated, the valve shall be capable of meeting all performance requirements specified in 3.8.

3.6.2.2 Reversibility. Direct acting and reverse acting actuators shall be interchangeable on body subassemblies without requiring modification or use of additional parts to the body subassemblies.

3.6.2.3 Yoke. Yoke construction shall provide for easy access to the stuffing box, stem connection, and spring adjuster from either side of the valve. Universal mounting pads shall be provided on opposite sides of the yoke for mounting valve positioners and other accessories provided.

3.6.2.4 Travel indicator. A travel indicator shall be provided to indicate valve plug position. The indicator scale shall be attached to the yoke and marked in inches on one side of a dividing line and the percentage of opening on the other.

3.6.2.5 Diaphragms. Diaphragms shall be deep molded and uniform contact shall be maintained between the diaphragm plate and diaphragm during full valve travel. Diaphragm shall be molded from Polychloroprene (Neoprene) or nitrile rubber (Buna-N) and shall be synthetic fabric reinforced to provide a minimum burst strength of 150 lb/in^2 over a 2 inch opening.

3.6.2.6 Diaphragm spring. The use of, only one diaphragm spring shall be required. The ends of the springs shall be ground within 1 degree of its perpendicular axis. The spring compression shall be readily adjustable while the valve is in service. Spring and spring parts shall be coated with a rust inhibiting material.

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3.6.2.7 Diaphragm case. The diaphragm case shall be suitable for a working gage pressure of 50 lb/in^2. The case shall be made of aluminum or steel coated with a rust inhibiting material.

3.6.2.8 Air connections. Air connections between the pilot, actuator, and other accessories shall be in accordance with MS16142 straight thread and O-ring seal.

3.7 Maintainability. Internal parts shall be readily accessible for easy disassembly and reassembly with standard tools and shall be configured to prevent, as far as practical, the incorrect reassembly of parts. Positioning and alinement of all parts in assembly shall employ positive means so that correct reassembly is repeatedly assured. In no case shall parts for a given valve be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance and strength.

3.7.1 Valve design shall permit accomplishment of the following maintenance actions within the time limits specified:

Action	Time allowed
Renew internal trim	3/4 hour
Remove, clean, and replace quiet flow regulating element	2-1/2 hours

3.7.2 Maintenance envelope. Valves shall be capable of being installed, operated, adjusted, and maintained within the envelope defined below:

Center line to top	33-1/2 inches
Center line to bottom	9-1/2 inches
Center line to side	7 inches

3.8 Performance requirements.

3.8.1 Performance modes. The performance requirements specified in 3.8.2 through 3.8.5 shall apply with use of a positioner as specified in 3.11.

3.8.2 Linearity/capacity. With inlet gage pressures within a range of 150 to 180 lb/in^2, inlet temperatures between saturation and 10°F superheat, and outlet gage pressures within a range of 0 to 30 lb/in^2, the relationship between air signal and stem travel in both directions shall be linear within plus or minus 3 percent (of full rated travel) over the full rated stroke of the valve. Flow control shall meet the points specified in table II without requiring any adjustment to the valve or controls during the specified sequence.

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TABLE II. Linearity/capacity requirements.

Stem travel (percent of full rated opening)	Inlet pressure (lb/in^2g)	Outlet pressure (lb/in^2g)	Flow (lb/hr) (plus or minus 10 percent)
0	150	0	0
12.5	150	1.5	687.5
25	150	3.5	1375
37.5	150	6	2062.5
50	150	10	2750
62.5	150	17	3437.5
75	150	28	4125
87.5	150	30	4812.5
100	150	30	5500
0	180	0	0
12.5	180	1.7	812.5
25	180	4	1625
37.5	180	8	2437.5
50	180	15	3250
62.5	180	27.5	4062.5
75	180	30	4875
87.5	180	30	5687.5
100	180	30	6500

3.8.3 Rangeability. Rangeability shall be at least 25:1. At 150 lb/in^2g inlet pressure, the valve shall be capable of smooth operation at flow rates as low as 220 pounds per hour (lb/hr).

3.8.4 Shut-off. The valve shall structurally withstand, but will not be required to shut off against, full design gage pressure of 850 lb/in^2. It shall, however, be capable of shut-off against an inlet gage pressure of 250 lb/in^2 (zero lb/in^2 outlet). Shut-off shall be defined as a flow not exceeding 10 lb/hr.

3.8.5 Seat tightness. Under normal operating conditions (as defined in 3.9), the valve shall meet the following seat tightness requirements: With a dead-end downstream volume not exceeding the volume represented by 100 diameters of insulated and untrapped downstream pipe, any steam leakage from the inlet to the outlet of the valve shall be less than or equal to a value which will cause a discharge pressure buildup of more than 10 lb/in^2 in a 1 hour period.

3.8.6 External leakage (valve body subassembly). There shall be no visible external leakage.

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3.8.7 Leakage (diaphragm case). There shall be no leakage past the diaphragm or externally from the diaphragm case under the test conditions as specified herein (see 4.5.4).

3.8.8 Packing leakage (stem). There shall be no visible leakage past the packing under normal operating conditions. Visible leakage is defined as an amount exceeding that which will form a fixed or stable meniscus or ring of water.

3.8.9 Controllability. Under any given set of operating conditions and at any position within the full stroke of the stem, a gradually applied change in signal pressure to the diaphragm actuator of no more than 0.50 lb/in^2 in either direction shall cause valve movement. Movement, once initiated, shall not exceed one percent of travel.

3.8.10 Linearity. With the stuffing box only handtight and the valve body at atmospheric pressure, the relationship between air pressure to the diaphragm actuator and stem travel shall be linear to within plus or minus 3 percent (of full rated travel).

3.8.11 Mechanical shock and vibration. Valve shall meet the mechanical shock requirements of MIL-S-901, grade A, class I and the environmental vibration requirements of MIL-STD-167-1, type I. This shall apply to the entire valve including all associated instrumentation. During and subsequent to these conditions, the equipment shall not fail to perform its principal function and the deviation in the instrument output pressure signal shall not exceed 3 percent of its output pressure range. Any instrumentation intended to be valve mounted should be so mounted during shock and vibration testing.

3.8.12 Noise. The valve shall meet the structureborne noise requirements of MIL-STD-740 for type 2 resiliently mounted equipment as measured on the valve body and connecting flanges, both parallel and perpendicular to the direction of flow. The valve shall be tested for noise at the operating points listed in table III. Inlet temperature shall be between saturation and 10°F superheat.

TABLE III. Noise test requirements.

Condition no.	Inlet gage pressure (lb/in^2)	Outlet gage pressure (lb/in^2)	Flow (lb/hr)
1	180	8.5	2500
2	180	15	3400
3	180	20	3700
4	180	25	3920
5	180	30	4200

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3.9 Service and ambient conditions. Valves shall comply with the following service and ambient conditions:

(a) Fluid - Steam
(b) Normal inlet pressure - 150 to 180 lb/in^{2}g
(c) Design pressure - 850 lb/in^{2}g
(d) Normal outlet pressures - 0 to 30 lb/in^{2}g
(e) Normal temperatures - Saturation to 530°F maximum
(f) Normal flow demand - 0 to 5,500 lb/hr at 150 lb/in^{2}g inlet pressure
(g) Shock - In accordance with grade A, class I of MIL-S-901
(h) Vibration - In accordance with type I of MIL-STD-167-1

3.10 Marking.

3.10.1 Body markings. The manufacturer's name or trademark and the body material composition shall be cast or forged integral with the valve body. The size, rating, and a flow arrow or "inlet" and "outlet" shall be cast or forged integral with the valve body or stamped on the outside diameter (o.d.) of the flanges.

3.10.2 Identification plates. Identification plates made of corrosion-resisting steel or brass shall be fastened to each valve and pilot. Identification plates shall be permanently fastened to a part of the valve not subjected to working pressure and which will not be covered by insulation. Identification plates shall contain or provide space for inclusion of the following data:

(a) Manufacturer's name
(b) MIL-V-24598
(c) ANSI 600 pound rated
(d) Manufacturer's model and part number or identification
(e) Manufacturer's drawing numbers (assembly and CD)
(f) Technical manual number
(g) 9-digit CID/APL number.

3.11 Instrumentation. The valve is controlled by a yoke-mounted positioner supplied with 40 lb/in^2 air pressure. Signal input to this positioner is 3 to 15 lb/in^2, generated by a control pilot. Both the pneumatic control pilot and the positioner will be supplied by the Government. The control pilot will be a Fisher Series 4160R-204, or equal, and the positioner will be a Moore Model 721P315(M-16)10830-21P29MI, or equal.

3.12 Technical data. The contractor shall prepare engineering drawings and a technical manual in accordance with the data ordering documents included in the contract or order (see 6.2.2) and as specified in 3.12.1 and 3.12.2.

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3.12.1 Drawings. In addition to the drawing content required by the data ordering document (see 6.2.2), the following technical features shall be included:

(a) Calculated average fluid velocity through various portions of the valve shall include the inlet and outlet ports and, at minimum, enough sections within the valve to cover any velocity change in excess of 25 percent.

(b) Construction details of the "quiet" throttling element, including: (1) data on maximum particulate sizes that can be passed without clogging, (2) brief discussion or rationale on the mechanics by which pressure breakdown is achieved without generating the noise level normally associated with orificial throttling, (3) method of fabrication: configuration and process of fabrication shall be described in sufficient detail to illustrate that adequate structural strength and freedom from flow passage obstruction is assured during production.

(c) Calculations verifying compliance with the flow passage and overall flow-rate consistency requirements specified in 3.6.1.8. Required dimensional inspections shall be detailed.

(d) Complete production details of the seat, disc, and stem assembly and all other replaceable internal trim.

(e) Layout of the pressure-containing envelope (body, bonnet, and bottom cap) giving dimensions which control compression of the spiral-wound gaskets. This is to assure that where remachining is necessary to repair the gasket-sealing surfaces on these parts, that compensating cuts can be accurately made to restore original gasket compression. This layout shall also specify the dimensional limits of such corrective remachining within which function of the valve remains unaffected.

(f) Recommended assembly torques, or equivalent procedures, for making up all joints and threaded assemblies.

(g) Tabulation of required gasket characteristics including all dimensions (with tolerances) and load versus compression characteristics (with tolerances).

(h) Mark areas to be radiographic, magnetic particle, or dye penetrant inspected.

3.12.2 Technical manual. In addition to the general requirements covered by the data ordering document (see 6.2.2) a table shall be included listing wrench sizes and assembly torques (or other equivalent procedures) for making up all joints and threaded assemblies.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system acceptable to the Government for supplies and services carried by this specification. The inspection system shall be in accordance with MIL-I-45208.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be conducted at a laboratory satisfactory to the Naval Sea Systems Command (NAVSEA). Drawings in accordance with 3.12.1 shall be submitted prior to authorization of first article tests. First article tests shall consist of the examination of 4.3.2 and tests specified in 4.5.1 through 4.5.11. Failure of valve material and function during tests shall be cause for rejection.

4.3.1 First article. The valve submitted for first article inspection shall be accompanied by the following items:

(a) Internal replaceable trim (quiet-flow regulating element, valve plug, seat ring, etc.) - 1 complete set
(b) Spiral wound gaskets and stem packing - 2 complete sets
(c) Instruction book for installation, operation, and maintenance - 1 complete set
(d) Drawings in accordance with 3.12.1 - 1 complete set

4.3.1.1 Accelerometer blocks. The first article inspection valve shall have two accelerometer blocks welded to the body at the approximate positions shown on figure 1. The accelerometer blocks on the body shall as closely as possible be centered over the region of exit flow from the quiet-flow regulating element. Blocks shall be a steel disc 1 inch in diameter and 3/8-inches thick; have a surface flatness of 700 micro-inches root mean square (rms) or better; and a surface

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finish of 125 micro-inches rms or better. The center of each block shall be drilled and tapped 10-32 x 1/4-inch deep minimum, perpendicular to the finished surfaces within plus or minus 1 degree. The blocks shall be preserved with a rust inhibiting coating.

4.3.1.2 Test instrumentation. The tests in 4.5.6, 4.5.7, and 4.5.11 shall be conducted using a positioner as specified in 3.11.

4.3.2 Examination prior to testing. Upon receipt of the first article it shall be disassembled, and visually and dimensionally examined, and hardness of guiding surfaces established, to determine conformance with the requirements of this specification and and valve drawings. On reassembly, the gaskets and stem packing shall be replaced with spares randomly selected from those provided with the valve. Any difficulty encountered in assembly or disassembly shall be noted. Accomplishment of the maintenance actions specified in 3.7.1 shall be demonstrated.

4.3.2.1 In lieu of disassembly for examination, valves may be visually and dimensionally examined prior to assembly.

4.4 Quality conformance inspection. Each production valve shall be examined as specified in 4.4.1, 4.4.1.1, and tested in accordance with 4.5.1 through 4.5.4.

4.4.1 Visual and dimensional examination. Each valve shall be visually and dimensionally examined to verify conformance with the requirements of this specification and the engineering drawings.

4.4.1.1 The quiet-flow regulating element shall have undergone, during production, the inspections detailed in 3.12.1(c). A complete record of the inspection results for each element shall be provided with the valve (or element).

4.5 Tests.

4.5.1 Nondestructive. Nondestructive tests shall be performed as specified in MIL-STD-278 and in accordance with MIL-STD-271. Acceptance criteria shall be in accordance with MIL-STD-278. This shall include radiographic, magnetic particle, and liquid penetrant inspection, pressure and visual testing.

4.5.2 Hydrostatic. The valve shall be pressurized with water at 2175 lb/in^2g to determine compliance with 3.8.6 of no external leakage, and to assure there is no permanent distortion or structural failure.

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4.5.3 Seat tightness. With 180 lb/in^{2}g steam at the valve inlet and zero lb/in^{2}g air pressure on the actuator diaphragm, the valve shall be checked for leakage in the outlet line to determine that leakage shall not exceed that specified in 3.8.5. During this test, a check shall be made for external leakage.

4.5.4 Diaphragm case leakage. A leakage test shall be made on each valve with air at a gage pressure of 50 lb/in^2 after complete assembly. Detection shall be by soap bubble solution. There shall be no visible leakage over a 5 minute period.

4.5.5 Shut-off. With 250 lb/in g^{2}steam at the valve inlet and zero lb/in^{2}g air pressure on the actuator diaphragm, the valve shall be checked for shut-off in accordance with 3.8.4. Flow shall not exceed 10 lb/hr.

4.5.6 Operational. The valve shall be operationally tested as specified in 4.5.6.1 through 4.5.6.3. The valve shall be subjected to the normal operating pressures and temperatures defined in 3.9 during these tests and stem leakage shall be checked to determine compliance with 3.8.8. Tapping of the valve, or other means to overcome sticking during the test, shall not be permitted.

4.5.6.1 Controllability. Controllability shall be tested at 10, 25, 50, 75, and 90 percent of stroke. A gradually applied change in pressure to the diaphragm actuator (i.e., if a valve positioner is supplied with the valve, it shall be bypassed during this test) of 0.50 lb/in^2 in either direction shall cause valve movement at each position in accordance with 3.8.9. Movement shall not exceed one percent of travel.

4.5.6.2 Linearity. Linearity of travel shall be tested at 0, 25, 50, 75, and 100 percent of stroke. With the valve body at atmospheric pressure, and the stuffing box only hand tight, the relationship between air pressure to the diaphragm actuator (i.e., if a valve positioner is supplied with the valve, it shall be bypassed for this test) and stem travel shall be linear as specified in 3.8.10. The valve shall operate smoothly with no overshoot.

4.5.6.3 Flow. The valve shall be flow and linearity tested at the points specified in table II. The valve shall operate smoothly with no overshoot. Flow shall fall within the range specified in table II for each condition specified. Inlet steam temperatures shall be between saturation and 10°F superheat.

4.5.7 Noise tests. The valve shall be noise tested to the criteria specified in 3.8.12.

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4.5.8 High-impact shock test. The valve shall be subjected to and meet the high-impact shock tests for grade A, class I of MIL-S-901 pressurized to 180 lb/in^{2}g with water (see 3.8.11).

4.5.9 Vibration test. Valve shall be vibration tested in accordance with type I of MIL-STD-167-1 pressurized to 180 lb/in^{2}g with water. There shall be no resonance in the range of frequency tested (see 3.8.11).

4.5.10 Post test examination. After completion of the tests specified in 4.5.1 through 4.5.9, the test valve shall be disassembled and visually and dimensionally examined to assure there is no damage, excessive wear, or signs of galling or pitting which shall be cause for rejection.

4.5.11 Endurance. The valve shall be operated for a total of 750 hours under shipboard conditions in accordance with 3.5. It shall operate at all times without evidence of instability, without deviating from the noise requirements of 3.8.12, and without requiring any maintenance, repair, or adjustment effort. At the successful completion of the 750 hours of operation, the valve shall be removed and completely disassembled. All parts shall be examined for signs of excessive wear or any other condition indicating impending failure or malfunction. Impending failure or malfunction is defined as one which can be expected to occur within 1 year of operation. Any such condition shall constitute grounds for failure of the valve to pass these tests, regardless of how satisfactorily the valve performed during the 750 hours of operation.

4.6 Inspection of preparation for delivery. The packaging, packing and marking shall be inspected for compliance with section 5 of this document.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation-packaging, packing and marking. Valves shall be individually preserved-packaged level A or C, packed level A, B, or C and marked in accordance with MIL-V-3, as specified (see 6.2.1).

5.1.1 Special marking. In addition to the marking, as applicable, required in 5.1 and 5.1.2, shelf life marking shall be applied. Shipping containers (see 5.1) shall be stenciled in accordance with the marking instructions of MIL-STD-740.

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5.1.2 Asbestos items. Item(s) containing asbestos/ asbestiform like materials shall be packaged individually, in sets, or quantity as required for use in a seated impermeable container. Flexible containers shall be heat-sealed, rigid containers shall be enclosed in a flexible container and heat-sealed. An asbestos caution label (see 6.2.1) shall be affixed to each interior (unit and intermediate package) and shipping container. The caution label shall state the following:

"CAUTION

CONTAINS ASBESTOS FIBERS. AVOID CREATING DUST. BREATHING ASBESTOS DUST MAY CAUSE SERIOUS BODILY HARM."

5.2 Cushioning and cushioning materials. Cushioning and cushioning materials, including the requirements for cushioning and dunnage (loose-fill) shall be in accordance with the requirements specified in MIL-R-196.

5.3 Technical manuals. Technical manuals, which accompany shipments that are packed level A or B, shall be packaged in a transparent waterproof plastic bag, minimum 4 mil thick. Closure shall be by heat sealing. Technical manuals shall not be placed within any flexible sealed barrier enclosing components. The copy(s) of the manual shall be placed in the shipping container housing the main unit. Packing lists shall indicate which container contains the technical manual(s) and shall also state the approximate location therein. For ease of removability the location of the manual shall be such that it is readily accessible when the container is opened. Technical manuals, when shipped in bulk quantities, shall not be individually wrapped, but shall be packed in accordance with the requirements of the applicable technical manual specification or packed in containers conforming to the requirements for level A, B, or C, as specified (see 6.2.1).

5.4 Data (see 3.12). Data shall be prepared for delivery in such a manner as to insure the required information is protected against deterioration, physical damage or loss during shipment from the contractor to the receiving activity. Packages or shipping containers shall as a minimum conform to the level C requirements of 5.1.

6. NOTES

6.1 Intended use. The quiet-design lithium bromide steam control valve specified herein is intended for service in submarine steam systems.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) If first article inspection is required (see 3.1, 4.3, and 6.3).
(c) Applicable levels of preservation-packaging and packing required (see 5.1).
(d) The caution labels shall conform to Occupational Safety and Health Administration (OSHA) Regulation Part 1910, Sections 145 and 1001, in accordance with Chief of Naval Operations, OPNAVINST 6260.1A (see 5.1.2).
(e) Use of polystyrene (loose-fill) material (see 5.2).
(f) Packing of technical manuals (see 5.3).

6.2.2 Data requirements. When this specification is used in a contract which invokes the provision of the "Requirements for Data" of the Defense Acquisition Regulation (DAR), the data identified below, which are required to be developed by the contractor, as specified on an approved Data Item Description (DD Form 1664), and which are required to be delivered to the Government, should be selected and specified on the approved Contract Data Requirement List (DD Form 1423) and incorporated in the contract. When the provisions of the "Requirements for Data" of the DAR are not invoked in a contract, the data required to be developed by the contractor and required to be delivered to the Government should be selected from the list below and specified in the contract.

Paragraph	Data requirements	Applicable DID	Option
3.12.1	Drawings, engineering and associated lists	DI-E-7031	Level II Design activity designation - Contractor Drawing No. - Contractor Delivery of hard copy - Contracting activity
3.12.2 and 5.3	Manual, technical preliminary	DI-M-2043	Type I of MIL-M-15071

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

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6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.2.2.2 Where the Government has limited rights in the data shown on the drawings, as determined by the contractual provisions regarding rights in technical data, the drawings may be marked with a legend. If used, the "Limited Rights Legend" of DAR should be used.

6.3 First article inspection. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
Navy - SH
(Project 4820-N400)

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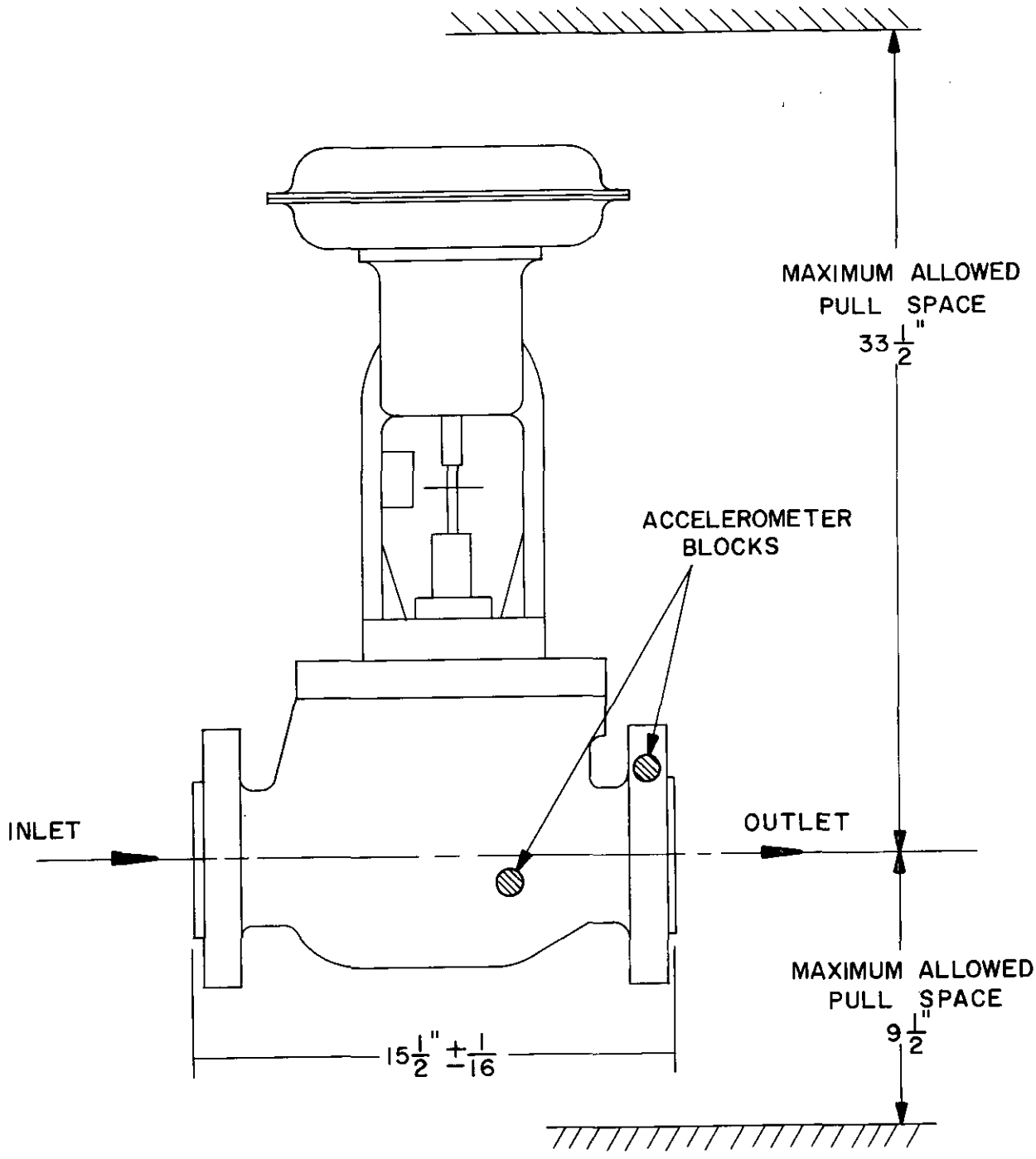


FIGURE 1. Interface dimensions.

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COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 3112)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
NAVY
DOD 316

U.S.MAIL

COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 3112)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362

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DOCUMENT IDENTIFIER AND TITLE
MIL-V-24598(SH)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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